

Madhav Anand Menon

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EDUCATION

University of Illinois Urbana-Champaign

Expected May 2028

B.S. Computer Science and Physics; Minor in Mathematics (James Scholar)

GPA: 3.97/4.0

- Yee Seung Ng Scholar; Harold and Ruth Hayward Scholarship; Dean's List; Tau Beta Pi Honours Society
- Relevant Coursework: Object Oriented Programming, Data Structures and Algorithms, Computer Architecture, Linear Algebra (Upcoming: Operating Systems, Numerical Methods, Databases)

EXPERIENCE

Disruption Lab

Feb. 2026 - Present

Software Engineer

Champaign, IL, USA

- Engineered a 5 stage Canvas-to-PostgreSQL RAG pipeline ingesting course content across 3 file formats (PDF/DOCX/TXT) into 1024-dim pgvector vector-database embeddings with 350 words sliding-window chunk overlap via AWS Bedrock
- Designed a prompt-engineered LLM skill extractor using Claude Haiku on AWS Bedrock with NACE framework alignment, returning deduplicated skill lists across multiple courses from a single API call
- Shipped a 9-endpoint FastAPI backend with custom HMAC-SHA256 auth (8-hr TTL, zero dependencies) and dual-mode PostgreSQL full-text and pgvector cosine similarity search

AlgoVerse AI

Sep. 2025 - May 2026

LLM Researcher

San Francisco, CA, USA (Remote)

- Built DeployStega, a covert differential-privacy LLM communication framework on GitHub using token-binning steganography that encodes up to 8 bits per word, achieving 90%+ word inclusion fidelity across a corpus of 13M+ events from 695K+ real users
- Modeled behavioral indistinguishability from empirical GH Archive priors (median inter-event gap: 179s; commit→commit transition probability: 0.899; commits = 78.5% of events) to constrain synthetic covert traces within a statistically normal feasibility region
- Evaluated detection resistance across 5 adversarial classifiers (logistic regression, random forest, SVM, TS-CNN, BERT-base) on 4000 synthetic users, using a differential-privacy-inspired ϵ metric derived from classifier TPR/FPR to quantify population-level indistinguishability

DigiAlert

Jun. 2025 - Aug. 2025

Software Engineer Intern

Santa Clara, CA, USA (Remote)

- Developed Flask-based AWS CSPM to visualise security metrics from 4 AWS services using Chart.js—reduced CSPM costs by 45%
- Engineered data pipeline using boto3, threading, and subprocess management to fetch, process, and cache security metrics and 500+ CloudTrail events (per refresh) with real-time CIS v5.0.0-based risk classification
- Used REST APIs and nginx/gunicorn to integrate dashboard into internal system; enabled time-based invalidation cache management

PROJECTS

GPU-Accelerated Ray Tracer | *CUDA, C++*

- Engineered a real-time ray tracer in CUDA achieving a 40x speedup over a single-threaded CPU baseline, rendering scenes at 60+ FPS at 1080p on an NVIDIA RTX 3060
- Implemented a BVH acceleration structure that reduced ray-intersection tests by 80%, cutting average frame render time from 95ms to 14ms
- Optimized memory coalescing and warp utilization to raise kernel occupancy from 50% to 88%, enabling rendering of scenes with 1M+ triangles and 10M+ rays per frame

Computational Astrophysics Research Project | *CLASS, SciPy, Matplotlib, pandas*

- Simulated Λ CDM neutrinos and investigated their breaking point and free-streaming length using the Cosmic Linear Anisotropy Solving System
- Developed algorithm for data processing and fitting through Boltzmann-solvers reducing processing time by 15%
- Computed power-matter spectrum of a primordial universe and fitted transfer function to an error of 4%

SKILLS

Languages: C++; C; Python; Java; SQL; Verilog; MIPS Assembly

Libraries and Frameworks: CUDA; PyTorch; FastAPI; Flask; pandas; scikit-learn; NumPy; Matplotlib

Tools: NVIDIA Nsight; AWS; Azure; GCP; ; Git; TeX/LaTeX; Docker; Valgrind, Mathematica, MATLAB

Other Activities: CS 124 Project Manager; [TEDx Speaker](#); ACM Corporate Team Member